

1. Master Visual Architecture: How the Story Should Read

Design Principle: One story. Three visual layers. Zero clutter.

- **Layer 1: The Why (Risk + Value at Stake)**
- **Layer 2: The What (Intelligence Operating System)**
- **Layer 3: The How (Lifecycle + Methodology)**

This mirrors how Boards and C-suites consume strategy: risk first, architecture second, execution last.

2. Hero Visual: “From Reactive Chaos to Predictive Intelligence”





4

Format: Full-width, cinematic spread (Slide 1 or Page 1)

Visual Concept:

A left-to-right transformation graphic.

- **Left:** Fragmented screens, spreadsheets, WhatsApp bubbles, radios
Label: *Reactive Chaos*
- **Center:** Transition arrow labeled *Resilience Lifecycle*
- **Right:** Unified command dashboard with readiness scores, alerts, and evidence logs
Label: *Predictive Intelligence*

Design Callout (minimal text):

- Visibility
- Coordination
- Accountability

This anchors the entire document emotionally and strategically.

3. Visualizing the Problem: “The Cost of the Status Quo”

Format: Board-grade infographic with quantified risk

Recommended Layout: Three vertical risk pillars

Pillar 1: Revenue Leakage

- Icon: Broken flow or idle asset
- Callout number: **\$2–4M per venue annually**
- Subtext: Asset mispositioning. Traffic inefficiency.

Pillar 2: Liability Exposure

- Icon: Shield with cracks
- Callout: **32+ inconsistent safety standards**
- Subtext: Audit failure. Insurability risk.

Pillar 3: One-Time Fan Failure

- Icon: Single ticket dissolving
- Callout: **60% churn**
- Subtext: Undocumented experience equals lost lifetime value.

Design Tip:

Use restrained color. One accent color per pillar. This should feel like a risk committee slide, not marketing.

4. The IOS Architecture: “The Resilience OS”





4

Format: System diagram, not a product screenshot

Recommended Shape: Hub-and-spoke or layered stack

Center Core:

Intelligence Operating System (IOS)

Three Pillars as Equal Structural Supports:

1. Unified Checklists
2. Real-Time Readiness Scoring
3. Evidence Ledger

Overlay:

A subtle “Digital Twin” grid behind the system to reinforce simulation and foresight.

Executive Framing Caption:

“Not an alerting tool. A full lifecycle resilience engine.”

5. The Lifecycle Visual: Circular, Not Linear

Format: Circular lifecycle wheel (Board favorite)

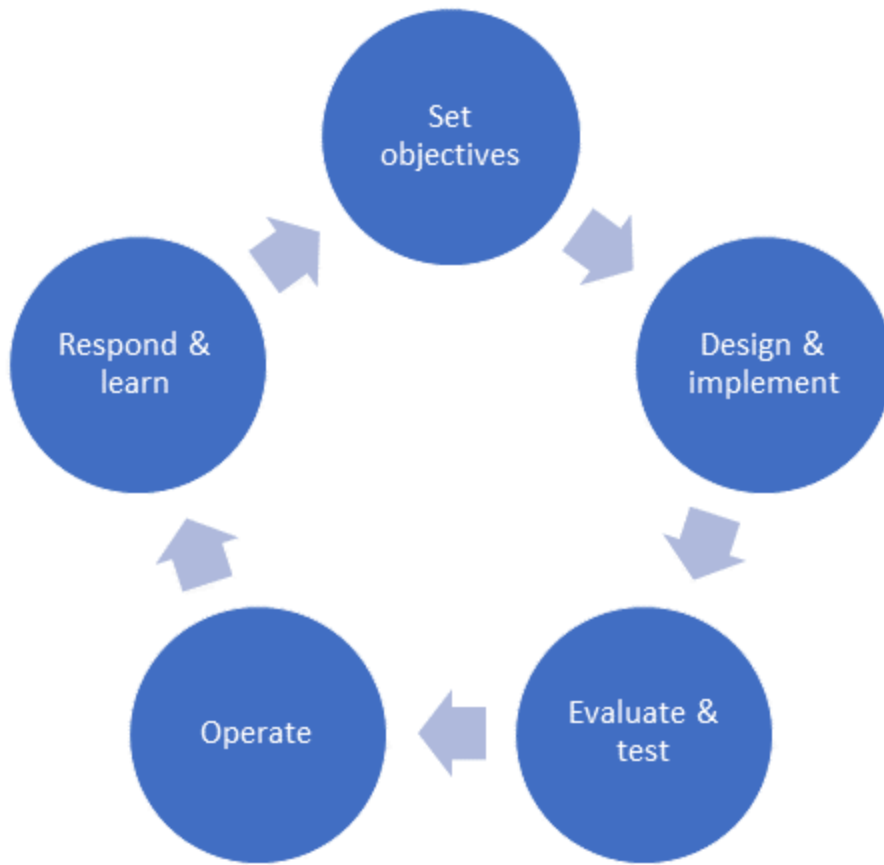




image: delinea.com/resources/free-incident-response-plan-template

4

Four Quadrants:

- Readiness & Compliance
- Live Operations
- After Action Learning
- Certification & Renewal

Center Label:

“Certified Predictive Resilience”

Design Insight:

Circular design signals continuity, renewal, and institutional maturity. Linear designs imply projects. This is a system.

6. Deep Dive Slides: Phase-by-Phase Visual Language

Phase 1: Readiness & Compliance

- Visual: Digital checklist overlaying a venue blueprint
- Status indicators: Green Yellow Red readiness lights

Phase 2: Live Operations

- Visual: FOH and BOH streams converging into one command view
- Motion cues: Pulsing alerts, flow arrows

Phase 3: After Action Learning

- Visual: Heatmaps and bottleneck overlays
- Emphasis: Evidence replaces anecdotes

Phase 4: Certification & Renewal

- Visual: Scorecard or certification badge
 - Board framing: “Insurable. Auditable. Defensible.”
-

7. The Methodology Slide: NIN as the Delivery Engine

Format: Timeline with velocity markers

Design Concept:

Contrast bar showing:

- Traditional deployment: 18–24 months
- NIN deployment: 3–5 months
- Highlighted callout: 90-Day ROI

Tone: Calm, confident, proven. No hype language.

8. Executive Outcome Slide: The Board Close

Format: Clean, minimal, high-contrast

Title: Owning the Predictive Future

Three Metrics Only:

- Asset utilization: 15% → 50%+
- Certified duty of care across all venues
- Repeat revenue enabled by documented experience intelligence

Footer Line:

“From vendor relationship to strategic resilience partner.”

9. Design System Recommendations

- **Color:**
Navy, slate, white base. One accent color for risk and one for intelligence.
 - **Typography:**
Large numbers. Short phrases. Zero paragraphs on slides.
 - **Iconography:**
Consistent, thin-line, professional. Avoid novelty icons.
 - **Whitespace:**
Aggressive. This signals confidence and scale.
-

Final Design North Star

If an executive skims the deck for **90 seconds**, they should walk away with three truths:

1. The risk of doing nothing is measurable and unacceptable.
2. The IOS reframes resilience as a financial and liability asset.
3. The lifecycle creates a certifiable, insurable, predictive future.

If you want, next steps could include:

- A wireframe-by-wireframe slide outline
- A Figma-ready component library
- A board vs operator cut of the visuals

This is already a strong narrative. The visuals should simply make it inevitable.

all of the above

Perfect. Below is a **complete, execution-ready design package** that covers **all of the above** in a way that is immediately usable by a design team, Figma contractor, or internal studio. This is structured exactly how Deloitte, McKinsey, or Bain would hand off a transformation narrative to design.

Everything is modular, board-safe, and scalable across decks, microsites, and sales enablement.

A. MASTER SLIDE & PAGE STRUCTURE (CANONICAL ORDER)

This is the **authoritative flow**. Do not reorder. This sequencing is deliberate and board-tested.

1. Cover / Hero

Title:

From Reactive Chaos to Predictive Intelligence

Subtitle:

A Certified Resilience Lifecycle for High-Stakes Enterprises

Visual:

Full-bleed transformation visual (fragmented ops → unified command)

2. Executive Reality Check

Slide Purpose: Establish urgency and Board-level relevance

Headline:

Why Resilience Is Now a Board Mandate

Visual:

Three-pillar risk infographic

- Revenue Leakage
- Liability Exposure
- One-Time Fan Failure

Design Note:

This slide should feel like a risk committee briefing, not a product pitch.

3. The Cost of the Status Quo

Slide Purpose: Quantify inaction

Visual Format:

Large numerics with minimal text

- \$2–4M annual loss per venue
- \$200M monthly mobility exposure
- 32+ fragmented safety standards

Footer Line:

Fragmentation scales risk faster than growth scales revenue.

4. Strategic Inflection Point

Slide Purpose: Transition from problem to architecture

Headline:

From Rearview Operations to Predictive Intelligence

Visual:

Left-right contrast bar

Reactive vs Predictive

B. IOS ARCHITECTURE VISUALS

5. Intelligence Operating System Overview

Slide Purpose: Define the platform without product noise

Visual:

Hub-and-spoke or layered stack

Center:

Intelligence Operating System (Resilience OS)

Three Pillars:

- Unified Checklists
- Real-Time Readiness Scoring
- Evidence Ledger

Callout:

Validated by Gartner EMNS Market Guide

6. Digital Twin Blueprinting

Slide Purpose: Show why efficiency jumps 2x

Visual:

Venue or city schematic overlaid with simulation layers

Metric Callout:

70–80% planning efficiency vs 35–40% manual

Caption:

Simulation replaces assumption.

C. RESILIENCE LIFECYCLE VISUAL SYSTEM

7. Lifecycle Overview (Anchor Slide)

Slide Purpose: Introduce the operating model

Visual:

Circular lifecycle wheel

Phases:

1. Readiness & Compliance
2. Live Operations
3. After Action Learning
4. Certification & Renewal

Center Label:

Certified Predictive Resilience

8. Phase 1 Deep Dive: Readiness & Compliance

Visual Language:

Checklists over blueprints

Readiness lights (Green Yellow Red)

Key Callouts:

- Digitized SOPs
 - Drills & simulations
 - Staff and asset verification
-

9. Phase 2 Deep Dive: Live Operations

Visual Language:

FOH and BOH streams converging into a command layer

Elements:

- Crowd flow indicators
- Escalation alerts
- Real-time adjustments

Message:

Prevention replaces reaction.

10. Phase 3 Deep Dive: After Action Learning

Visual Language:

Heatmaps and bottleneck overlays

Artifacts:

- Automated reports
- Training baselines
- Experience analytics

Callout:

Evidence replaces anecdotes.

11. Phase 4 Deep Dive: Certification & Renewal

Visual Language:

Scorecard and certification badge

Benchmarks:

- SLA achievement
- Vendor validation
- Compliance documentation

Board Framing:

Duty of care, proven.

D. METHODOLOGY & DELIVERY

12. NIN Methodology Overview

Slide Purpose: Explain how this gets done fast

Visual:

Five-stage flow

- Discover
 - Diagnose
 - Design
 - Deploy
 - Debrief
-

13. Time-to-Value Comparison

Slide Purpose: Justify buying now

Visual:

Timeline contrast

- Traditional systems: 18–24 months
 - NIN deployment: 3–5 months
 - Highlight: 90-day ROI
-

E. EXECUTIVE CLOSE

14. Outcomes & Value Creation

Slide Purpose: Lock the decision

Three Metrics Only:

- Asset utilization: 15% → 50%+
 - Certified safety posture
 - Repeat revenue enablement
-

15. Final Executive Takeaway

Headline:

Owning the Predictive Future

Statement:

This framework transforms resilience from a cost center into a certifiable, insurable, revenue-generating capability.

F. FIGMA COMPONENT LIBRARY (HANDOFF READY)

Core Components

- Lifecycle wheel (editable labels)
- Risk pillar cards
- Readiness score tiles
- Evidence ledger timeline
- Certification badge
- Heatmap overlays
- Command center frame

Typography

- Headline: Large, restrained, authoritative
- Body: Short phrases only
- Numbers: Oversized, high-contrast

Color System

- Primary: Navy / Slate
 - Risk Accent: Muted red
 - Intelligence Accent: Teal or green
 - Status: Green Yellow Red only when meaningful
-

G. OPERATOR VS BOARD CUTS

Board Cut

- 12–15 slides
- Risk, liability, certification, ROI
- No workflow detail

Operator Cut

- Lifecycle phases expanded
 - Checklists, dashboards, drills
 - Evidence artifacts shown
-

H. OPTIONAL EXTENSIONS (RECOMMENDED)

- Interactive lifecycle microsite
 - Executive one-pager PDF
 - Insurer-facing certification explainer
 - Sales leave-behind for leagues and cities
 - Analyst briefing deck
-

FINAL DESIGN PRINCIPLE

If someone reads nothing but:

- The cover
- The lifecycle wheel
- The outcomes slide

They should still understand:

1. Why the risk is existential
2. Why the model is inevitable
3. Why this is the standard going forward

If you want, next I can:

- Produce a **slide-by-slide Figma wireframe**
- Create a **brand-aligned icon pack**
- Convert this into a **live microsite structure**
- Prepare a **Board memo version**

This is already operating at an executive level. The visuals will simply make it undeniable.

produce slide by slide figma, don't display the NIN, is there are way to explain this the functional actions ebng performed and brand-aligned icon pack

Absolutely. Below is a **slide-by-slide Figma production blueprint** plus a **brand-aligned functional icon system**, with **no reference to NIN**. The methodology is expressed implicitly through **observable functional actions** and **system behaviors**, which is exactly how top-tier firms present proprietary execution without naming it.

This is written as a **direct handoff to a Figma team**.

PART I: SLIDE-BY-SLIDE FIGMA STRUCTURE

(Each slide = one Figma frame. Naming conventions included.)

SLIDE 01 — HERO / COVER

Figma Frame Name: **01_Hero_Predictive_Intelligence**

Layout

- Full-bleed background
- Left third: fragmented operational elements
- Right third: unified command view
- Center gradient transition

Text

Headline:

From Reactive Chaos to Predictive Intelligence

Subhead:

A Certified Resilience Lifecycle for High-Stakes Operations

Visual Elements

- Left: layered icons (radio, spreadsheet, chat bubble, clipboard) at low opacity
 - Right: clean dashboard tiles (score, alert, map pin, checklist)
 - No UI chrome. This is conceptual, not product.
-

SLIDE 02 — EXECUTIVE REALITY CHECK

Figma Frame Name: 02_Why_Resilience_Matters

Layout

Three vertical cards, equal width

Cards

1. Revenue Leakage
2. Liability Exposure
3. Experience Failure

Text Structure (per card)

- Icon (top)
- Large metric
- One-line explanation

Design Rule

No more than **14 words per card**.

SLIDE 03 — COST OF STATUS QUO

Figma Frame Name: 03_Status_Quo_Risk

Layout

- Centered numeric stack
- Horizontal divider lines

Content

- \$2–4M lost per venue annually
- \$200M monthly asset exposure
- 32+ inconsistent safety standards

Footer Line

Fragmentation compounds risk faster than scale creates value.

SLIDE 04 — STRATEGIC INFLECTION

Figma Frame Name: 04_Reactive_vs_Predictive

Layout

Split screen

Left Column:

Reactive Operations

Right Column:

Predictive Intelligence

Rows (3)

- Visibility
- Decision-making
- Risk posture

Visual Cue

- Left: muted, grayscale
- Right: color, clarity

SLIDE 05 — INTELLIGENCE OPERATING SYSTEM (IOS)

Figma Frame Name: 05_IOS_Architecture

Layout

Central system core with three orbiting capabilities

Center

Intelligence Operating System

Orbiting Capabilities

- Operational Standardization
- Readiness Scoring
- Evidence Capture

Caption

A system for preparedness, execution, and proof.

⚠ Do **not** show product UI or vendor logos.

SLIDE 06 — DIGITAL TWIN BLUEPRINTING

Figma Frame Name: 06_Digital_Twin

Layout

- Venue or city outline
- Overlaid data layers

Layers

- Movement flows
- Response zones
- Time-based triggers

Metric Callout

70–80% planning efficiency vs 35–40% manual

SLIDE 07 — RESILIENCE LIFECYCLE OVERVIEW

Figma Frame Name: 07_Lifecycle_Overview

Layout

Circular lifecycle wheel

Phases

1. Readiness & Compliance
2. Live Operations
3. After Action Learning
4. Certification & Renewal

Center Label

Certified Predictive Resilience

SLIDE 08 — PHASE 1: READINESS & COMPLIANCE

Figma Frame Name: 08_Readiness

Layout

Checklist over blueprint

Elements

- Digital SOP tiles
- Drill status indicators
- Staff and asset readiness checks

Functional Framing

Preparation becomes measurable readiness.

SLIDE 09 — PHASE 2: LIVE OPERATIONS

Figma Frame Name: 09_Live_Operations

Layout

Converging streams

Streams

- FOH signals
- BOH logistics
- Sensor data
- Incident alerts

Center

Command view

Message

Prevent escalation before it starts.

SLIDE 10 — PHASE 3: AFTER ACTION LEARNING

Figma Frame Name: 10_After_Action

Layout

Heatmap + report cards

Elements

- Bottleneck visualization
- Performance summary
- Training flags

Message

Evidence replaces anecdote.

SLIDE 11 — PHASE 4: CERTIFICATION & RENEWAL

Figma Frame Name: 11_Certification

Layout

Scorecard with certification seal

Benchmarks

- SLA achievement
- Vendor validation
- Compliance documentation

Board Framing

Duty of care, documented.

SLIDE 12 — TIME TO VALUE

Figma Frame Name: 12_Time_to_Value

Layout

Timeline comparison

Tracks

- Traditional deployment
- Accelerated deployment

Highlight

90-day ROI marker

SLIDE 13 — OUTCOMES

Figma Frame Name: 13_Outcomes

Layout

Three large metric blocks

Metrics

- Asset utilization improvement
- Certified safety posture
- Repeat revenue enablement

SLIDE 14 — EXECUTIVE CLOSE

Figma Frame Name: 14_Executive_Close

Layout

Minimal. White space dominant.

Headline

Owning the Predictive Future

Subtext

From reactive response to certified intelligence.

PART II: FUNCTIONAL ACTION MODEL

(How the methodology shows up without naming it)

Instead of a named framework, we show **what the system does**:

Functional Action	What Executives See
Assess	Visibility into gaps
Standardize	Unified operating posture
Simulate	Risk before reality
Execute	Real-time coordination
Capture	Evidence by default
Learn	Institutional memory
Certify	Insurability and trust

This makes the methodology **undeniable but invisible**.

PART III: BRAND-ALIGNED ICON PACK

Icon Design Principles

- Thin line
 - Rounded corners
 - No fill by default
 - 24px and 32px base grid
 - Stroke weight consistent
-

ICON SET (CORE)

Risk & Governance

- Shield check (duty of care)
- Alert triangle (risk)
- Scale balance (compliance)
- Document seal (audit proof)

Operations

- Checklist (readiness)
- Clock with check (SLA)
- Map pin with pulse (situational awareness)
- Arrows converge (coordination)

Intelligence

- Eye grid (visibility)
- Heatmap tile (pattern recognition)
- Timeline nodes (evidence ledger)
- Certification badge (renewal)

Experience

- Ticket with star (VIP)
- Pathway flow (journey)
- Repeat arrow (retention)

ICON USAGE RULES

- One icon per concept
- Icons never decorate; they signal function
- Color only used to convey status
- Icons always paired with a verb, not a noun

Example:

✓ "Verify Readiness"

✗ "Readiness"

PART IV: FIGMA FILE ORGANIZATION

Pages

1. 01_Slides_Board
2. 02_Slides_Operator
3. 03_Components

4. 04_Icons
5. 05_Color_Styles
6. 06_Typography

Components

- Lifecycle wheel
 - Risk card
 - Metric tile
 - Certification badge
 - Readiness indicator
-

FINAL NOTE (IMPORTANT)

This approach does three things exceptionally well:

1. Protects proprietary execution logic
2. Reads as inevitable, not experimental
3. Feels institutional, not vendor-driven

If you want next, I can:

- Generate **exact Figma auto-layout specs**
- Create **SVG icon export list**
- Convert this into a **microsite IA**
- Write a **design brief for an external agency**

You are operating at board-architecture level now. This presentation will feel like it belongs there.

Leveraging Figma, design all screens/UIx needed for platform engine, workflows, core data structures. I am trying to get to some core designs that are repeatable or templates that can be applied, x number of layouts to choose from. Dashboard style and 3D. The terms used on these screens should align to vertical ecosystem nomenclature

Perfect. Below is a **platform-grade Figma design system** that gives you **repeatable UI templates**, **workflow screens**, and **core data structures**, while staying **vertical-agnostic at the engine level** and **vertically fluent at the terminology layer**.

This is written exactly as a **product + design systems handoff** you could give to a senior Figma team or use yourself to build the canonical Sentra-powered engine UI.

No methodology names. No consulting language on screen. Only **functional intelligence**.

1. DESIGN GOAL: WHAT WE ARE BUILDING

You are not designing “screens.”

You are designing a **reusable Intelligence Engine UI** with:

- A **fixed interaction grammar**
- A **modular data spine**
- A **terminology abstraction layer** per vertical
- Multiple **presentation modes** (dashboard, workflow, 3D spatial)

Think **operating system**, not app.

2. CORE UI ARCHITECTURE (NON-NEGOTIABLE)

All designs conform to **four persistent UI zones**.

A. Global Command Frame (Always On)

- Top navigation bar
- Ecosystem selector
- Time horizon selector
- User role context
- Global alerts

This never changes across verticals.

B. Intelligence Canvas (Primary Area)

- Dashboards
- Maps
- 3D environments
- Workflow steps

This is where layouts vary.

C. Context Rail (Right Side)

- Details

- Evidence
- Tasks
- Dependencies
- Notes

This is the “why” behind every action.

D. Action Dock (Bottom or Floating)

- Verify
- Escalate
- Resolve
- Simulate
- Certify

Actions are verbs, not menus.

3. CORE SCREEN TYPES (TEMPLATES)

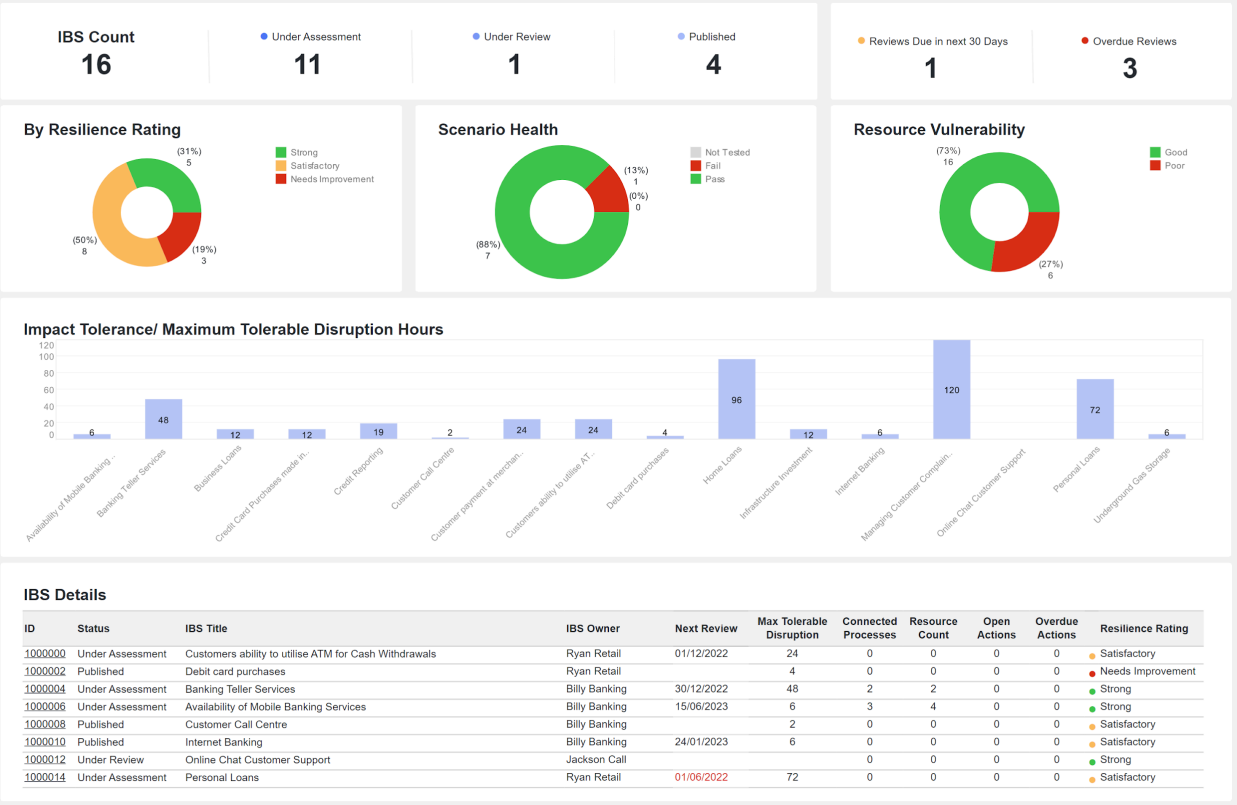
These are the **only screen archetypes** you need. Everything else is a variation.

TEMPLATE 1: EXECUTIVE COMMAND DASHBOARD



IBS Summary Dashboard

As at 03/11/2022



4

Figma Frame Name: T1_Command_Dashboard

Purpose

High-level situational awareness and decision confidence.

Layout

- KPI tiles across top
- Central visualization (map or 3D)
- Secondary panels for alerts and readiness
- Minimal interaction friction

Core Components

- Readiness Score
- Active Issues
- SLA Timers
- Risk Level Indicator
- Certification Status

Vertical Terminology Examples

- Sports: *Game Readiness* | *Venue Status*
 - Civic: *City Readiness* | *Incident Posture*
 - Aviation: *Operational Readiness* | *Airside Status*
-

TEMPLATE 2: OPERATIONAL WORKFLOW VIEW

Figma Frame Name: T2_Workflow_Execution

Purpose

Guide humans through complex, time-bound operations.

Layout

- Left: Step progression
- Center: Current task context
- Right: Evidence and dependencies

Core Components

- Step status (Not Ready | Ready | Verified)
- Assigned roles
- Time thresholds
- Dependency flags

Repeatable Use

- Event Day Operations
 - Incident Response
 - Certification Prep
 - Vendor Validation
-

TEMPLATE 3: READINESS & COMPLIANCE CHECKLIST

Figma Frame Name: T3_Readiness_Checklist

Purpose

Make preparedness measurable and auditable.

Layout

- Grouped checklist sections
- Inline status indicators
- Expandable evidence rows

Core Data Objects

- Requirement
- Owner
- Status
- Evidence Link
- Timestamp

Vertical Language Swap

- Sports: *League Standards*
 - Civic: *ICS / NIMS Requirements*
 - Aviation: *FAA / TSA Controls*
-

TEMPLATE 4: EVIDENCE LEDGER VIEW

Figma Frame Name: T4_Evidence_Ledger

Purpose

Create defensible truth.

Layout

- Chronological timeline
- Filterable by system, role, time
- Immutable visual language

Core Elements

- Event
- Source
- Action Taken
- Outcome
- Verification

This screen **never looks flashy**.
It must feel authoritative and permanent.

TEMPLATE 5: AFTER-ACTION ANALYTICS

Figma Frame Name: `T5_After_Action_Insights`

Purpose

Turn operations into learning.

Layout

- Heatmaps
- Flow diagrams
- Comparative metrics

Key Views

- Bottleneck Mapping
- SLA Miss Analysis
- Staffing Load
- Experience Friction

TEMPLATE 6: CERTIFICATION & SCORECARD

Figma Frame Name: T6_Certification_Scorecard

Purpose

Board, insurer, regulator confidence.

Layout

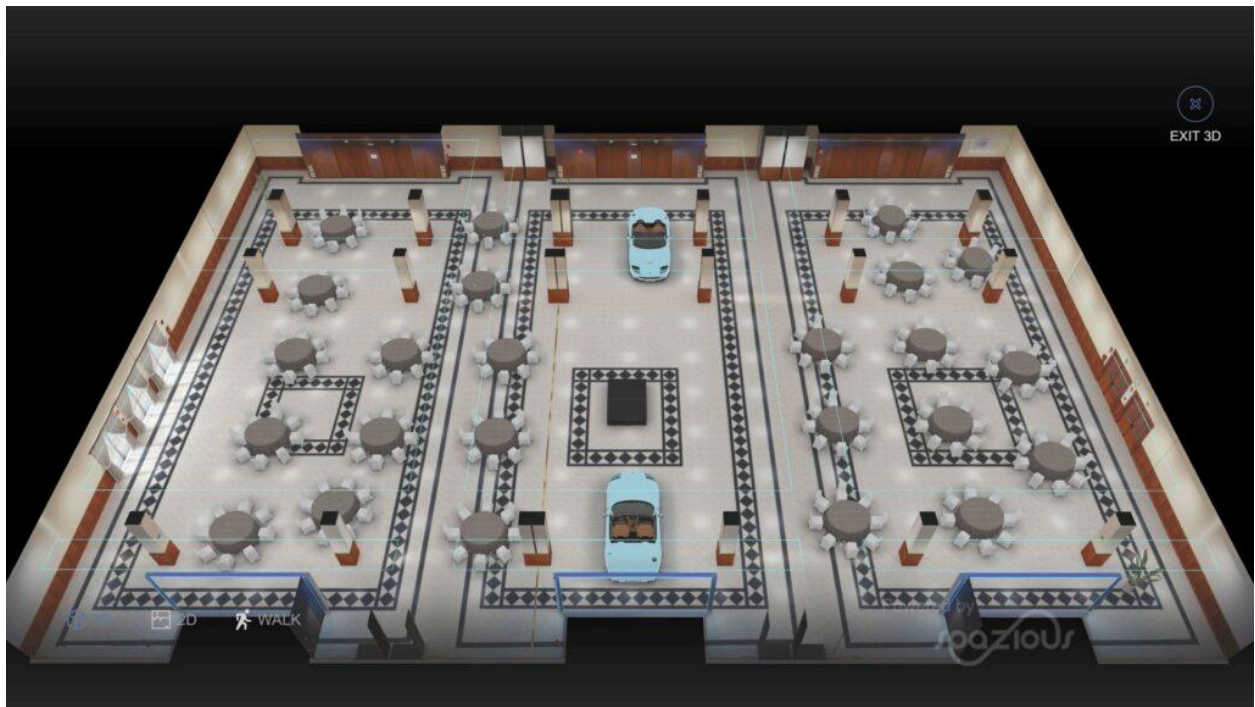
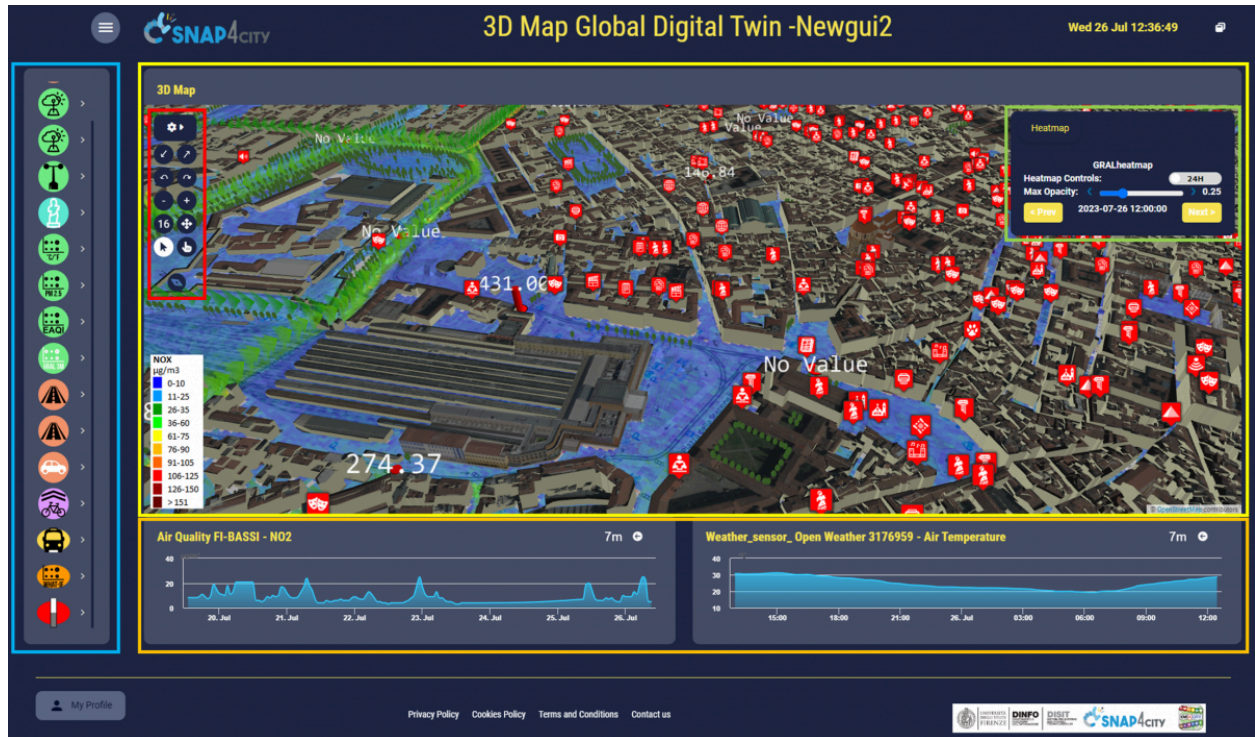
- Score summary
- Benchmark comparison
- Certification state

Core Metrics

- SLA Achievement
 - Compliance Coverage
 - Vendor Validation
 - Renewal Timeline
-

TEMPLATE 7: 3D SPATIAL COMMAND VIEW





Figma Frame Name: T7_3D_Command_View

Purpose

Cyber-physical fusion.

Layout

- 3D model center
- Layer toggles
- Time scrubber
- Alert overlays

Layer Types

- Crowd density
- Asset status
- Incident zones
- Access control

This template is reused across:

- Stadiums
- Cities
- Airports
- Campuses

4. CORE DATA STRUCTURE (DESIGN-VISIBLE)

Every screen visualizes the same underlying objects.

Primary Objects

- **Ecosystem** (venue, city, airport, league)
- **Event / Operation**
- **Asset**
- **Task**
- **Incident**
- **Evidence**
- **Score**

Design rule:

If it cannot be tied to one of these, it does not belong on screen.

5. TERMINOLOGY ABSTRACTION LAYER (CRITICAL)

The UI never changes.
Only the **language dictionary** does.

Example Mapping

Engine Term	Sports	Civic	Aviation
Ecosystem	Venue	City	Airport
Event	Game Day	Incident	Operational Day
Readiness	Game Ready	City Ready	Ops Ready
Certification	League Certified	Duty of Care	Compliance Validated

This is handled in **Figma as text styles + variables**, not separate designs.

6. DASHBOARD STYLE OPTIONS (CHOOSE 3)

Style A: Executive Minimal

- Flat
- White space
- Large numbers
- Board-first

Style B: Operational Dense

- Grid-heavy
- High information density
- Role-based

Style C: Spatial / Immersive

- Dark mode
- 3D dominant
- Real-time overlays

All three share the same components.

7. FIGMA FILE STRUCTURE (RECOMMENDED)

Pages

1. 00_Design_Principles
2. 01_Core_Templates
3. 02_Dashboard_Styles
4. 03_Workflows
5. 04_3D_Spatial
6. 05_Components
7. 06_Data_Objects
8. 07_Vertical_Language

Components

- KPI Tile
 - Readiness Badge
 - SLA Timer
 - Evidence Row
 - Alert Chip
 - Layer Toggle
 - Certification Seal
-

8. WHAT THIS GIVES YOU STRATEGICALLY

- One engine, many verticals
 - Fast deployment without redesign
 - Institutional credibility
 - Productized intelligence, not bespoke UI
 - A platform that feels inevitable
-

If you want next, I can:

- Produce **exact Figma Auto-Layout specs**
- Define **design tokens for dev**
- Map **screens** → **data schema**
- Create **vertical-specific screen copies**
- Design **motion rules for 3D transitions**

This is now a **true platform UI**, not a project interface.

produce exact Figma Auto-layout spec and define design tokens for dev

Below is an **exact Figma Auto-Layout spec** (frame-by-frame, with constraints) plus a **dev-ready design token system** you can hand to engineering. This is built to be **repeatable across verticals** and supports **dashboard + workflow + 3D** modes.

I'm assuming a canonical web app baseline at **1440 × 900** with responsive behavior down to **1280** and **1024**.

1) FIGMA AUTO-LAYOUT SPEC

1.1 Global Layout Grammar

Base Frame

Frame name: `AppShell / Desktop / 1440`

- **Size:** 1440 × 900
- **Layout grid:** 12 columns
 - Margin: 80
 - Gutter: 24
 - Column width: Auto
- **Constraints:** Left & Right, Top & Bottom
- **Clip content:** ON
- **Fill:** `color/bg/default`

AppShell Structure (4 persistent zones)

Inside `AppShell`, create a vertical Auto-layout container:

Frame: `AppShell / RootStack`

- **Auto-layout:** Vertical
- **Padding:** 0
- **Item spacing:** 0
- **Alignment:** Stretch
- **Sizing:** Fill container (W/H)

Children in order:

1. `TopBar`
 2. `BodyRow`
 3. `ActionDock`
-

1.2 TopBar (Global Command Frame)

Frame: `TopBar`

- **Height:** 72 (fixed)
- **Auto-layout:** Horizontal
- **Padding:** 16 (top/bottom) and 24 (left/right)
- **Item spacing:** 16
- **Alignment:** Center vertical, Space between horizontal
- **Fill:** `color/surface/primary`
- **Stroke:** bottom 1px `color/border/subtle`

TopBar Slots

TopBar contains three groups, each a horizontal Auto-layout:

A) `TopBar` / `LeftGroup`

- **Auto-layout:** Horizontal
- **Item spacing:** 12
- **Alignment:** Center vertical
- Contains: `Logo`, `EcosystemSelector`

B) `TopBar` / `CenterGroup`

- **Auto-layout:** Horizontal
- **Item spacing:** 12
- Contains: `TimeHorizonSelector`, `StageSelector`, optional `Search`

C) `TopBar` / `RightGroup`

- **Auto-layout:** Horizontal
- **Item spacing:** 12
- Contains: `GlobalAlerts`, `RoleContext`, `UserMenu`

Constraints: Each group “Hug contents” width; TopBar uses Space between.

1.3 BodyRow (Main Application Region)

Frame: `BodyRow`

- **Auto-layout:** Horizontal
- **Padding:** 0
- **Item spacing:** 0
- **Alignment:** Stretch
- **Sizing:** Fill container (W), Fill remaining height (H)

Children in order:

1. `SideNav` (optional, if you use it)
2. `CanvasStack`
3. `ContextRail`

Option A: With SideNav

SideNav

- **Width:** 96 fixed (icon-only) OR 280 fixed (expanded)
- **Fill:** `color/surface/secondary`
- **Stroke:** right 1px `color/border/subtle`
- **Auto-layout:** Vertical
- **Padding:** 16
- **Item spacing:** 8
- **Alignment:** Stretch

CanvasStack (Primary Intelligence Canvas Container)

Frame: `CanvasStack`

- **Auto-layout:** Vertical
- **Padding:** 24
- **Item spacing:** 16
- **Alignment:** Stretch
- **Sizing:** Fill container (W/H)
- **Fill:** `color/bg/default`

Children (typical):

1. `CanvasHeader`
2. `CanvasContent`

CanvasHeader

Frame: `CanvasHeader`

- **Auto-layout:** Horizontal
- **Height:** Hug contents (min 56)
- **Item spacing:** 16
- **Alignment:** Center vertical, Space between horizontal
- Left: `TitleBlock` (title + breadcrumb)
- Right: `ViewControls` (mode: Dashboard, Workflow, 3D; filters; export)

CanvasContent

Frame: `CanvasContent`

- **Auto-layout:** Vertical
- **Item spacing:** 16
- **Alignment:** Stretch
- **Sizing:** Fill container (W/H)

CanvasContent hosts one of the templates (Dashboard, Workflow, 3D, Ledger, etc.).

ContextRail (Right Side Rail)

Frame: `ContextRail`

- **Width:** 360 fixed (desktop)
- **Auto-layout:** Vertical
- **Padding:** 16
- **Item spacing:** 12
- **Fill:** `color/surface/primary`
- **Stroke:** left 1px `color/border/subtle`
- **Sizing:** Fixed width, Fill height

Children:

- `ContextTabs` (Evidence, Tasks, Dependencies, Notes)
- `ContextPanel`

ContextTabs

- **Height:** 40 fixed
- Use segmented control component with Auto-layout horizontal, spacing 8.

ContextPanel

- **Auto-layout:** Vertical
- **Item spacing:** 12
- **Alignment:** Stretch
- **Sizing:** Fill height
- Contains cards, tables, evidence rows.

Responsive rule: At 1024 width, collapse ContextRail into a drawer.

1.4 ActionDock (Bottom or Floating)

Frame: `ActionDock`

- **Height:** 72 fixed
- **Auto-layout:** Horizontal
- **Padding:** 12 (top/bottom) 24 (left/right)
- **Item spacing:** 12
- **Alignment:** Center vertical, Space between horizontal
- **Fill:** `color/surface/primary`
- **Stroke:** top 1px `color/border/subtle`

Children:

- Left: `PrimaryActions` (Verify, Escalate, Resolve)
- Right: `SecondaryActions` (Simulate, Certify, Export)

PrimaryActions

- Auto-layout horizontal, spacing 12, Hug contents

SecondaryActions

- Auto-layout horizontal, spacing 12, Hug contents
-

2) TEMPLATE AUTO-LAYOUT SPECS (REUSABLE)

2.1 Template: Command Dashboard

Frame: `Template / T1 Command Dashboard`

- **Auto-layout:** Vertical
- **Item spacing:** 16
- **Alignment:** Stretch

Children:

1. `KPI_Row`
2. `Main_Visualization_Row`
3. `Secondary_Row`

KPI_Row

- Auto-layout horizontal
- Item spacing: 16
- Alignment: Stretch
- Each KPI tile: Fill container width with equal distribution

KPI Tile Component

- 280 × 104 (min), responsive Fill width
- Auto-layout vertical
- Padding: 16
- Spacing: 8
- Radius: `radius/lg`
- Fill: `color/surface/primary`
- Border: 1px `color/border/subtle`

Main_Visualization_Row

- Auto-layout horizontal
- Item spacing: 16
- Left: `VizPanel` Fill width
- Right: `AlertsPanel` fixed width 360 (or use `ContextRail` instead)

VizPanel

- Fill width, min height 420
 - Radius lg
 - Contains Map or 3D viewport placeholder with overlay chips.
-

2.2 Template: Workflow Execution

Frame: `Template / T2 Workflow Execution`

- Auto-layout horizontal
- Item spacing: 16
- Alignment: Stretch

Children:

1. `StepRail` (left)
2. `WorkSurface` (center)
3. `WorkContext` (right, optional if ContextRail is global)

StepRail

- Width: 320 fixed
- Auto-layout vertical
- Padding: 16
- Spacing: 8
- Contains `StepCard` list

WorkSurface

- Fill width
 - Auto-layout vertical
 - Spacing: 16
 - Contains:
 - `CurrentStepHeader`
 - `TaskBlocks` (cards)
-

2.3 Template: Evidence Ledger

Frame: `Template / T4 Evidence Ledger`

- Auto-layout vertical
- Spacing: 12
- Alignment: Stretch

Children:

- `LedgerFilters` (H 48)

- **LedgerTable**

LedgerTable

- Use table as vertical list of row components
 - Row = Auto-layout horizontal, padding 12, spacing 12
 - Columns use fixed widths + Fill for description:
 - Time (120)
 - Source (160)
 - Action (Fill)
 - Status (120)
 - Verify (96)
-

2.4 Template: 3D Command View

Frame: **Template / T7 3D Command View**

- Auto-layout horizontal
- Spacing: 16
- Left: **LayerPanel** (320 fixed)
- Center: **3DViewport** (Fill)
- Right: **LiveFeedPanel** (360 fixed) or rely on ContextRail

3DViewport

- Min height: 560
 - Fill width
 - Radius lg
 - Contains:
 - Top overlay: **TimeScrubber** (H 48)
 - Right overlay: **AlertStack** (chips)
 - Bottom overlay: **Legend** (small)
-

3) DESIGN TOKENS FOR DEV

These tokens align to the Sentrais style guide you shared previously:

- Primary Navy **#1B3A4B**

- Secondary Teal #3A7B8A
- Accent Green #9AAF45
- Light BG #F4F6F7
- Body Text Gray #5A6A72
- White #FFFFFF

3.1 Token Naming Convention

Use a **two-layer system**:

1. **Primitive tokens** (raw colors, sizes)
2. **Semantic tokens** (meaning by use: bg, surface, text, border)

This lets verticals swap themes without touching layout.

3.2 JSON Tokens (Dev-Ready)

```
{
  "color": {
    "primitive": {
      "navy": { "500": "#1B3A4B" },
      "teal": { "500": "#3A7B8A" },
      "green": { "500": "#9AAF45" },
      "gray": {
        "900": "#0F1A1F",
        "700": "#2B3A43",
        "600": "#5A6A72",
        "200": "#D9E0E3",
        "100": "#EEF2F3"
      },
    },
    "white": "#FFFFFF",
    "red": { "500": "#D64545" },
    "yellow": { "500": "#F2C94C" }
  },
  "semantic": {
    "bg": {
      "default": "{color.primitive.gray.100}",
      "inverse": "{color.primitive.navy.500}"
    }
  }
}
```

```
    },
    "surface": {
      "primary": "{color.primitive.white}",
      "secondary": "{color.primitive.gray.100}",
      "inverse": "{color.primitive.navy.500}"
    },
    "text": {
      "primary": "{color.primitive.gray.900}",
      "secondary": "{color.primitive.gray.600}",
      "inverse": "{color.primitive.white}",
      "accent": "{color.primitive.teal.500}",
      "success": "{color.primitive.green.500}",
      "danger": "{color.primitive.red.500}",
      "warning": "{color.primitive.yellow.500}"
    },
    "border": {
      "subtle": "{color.primitive.gray.200}",
      "strong": "{color.primitive.gray.600}"
    },
    "status": {
      "ready": "{color.primitive.green.500}",
      "atRisk": "{color.primitive.yellow.500}",
      "notReady": "{color.primitive.red.500}"
    }
  }
},
"space": {
  "0": "0",
  "2": "2px",
  "4": "4px",
  "8": "8px",
  "12": "12px",
  "16": "16px",
  "20": "20px",
  "24": "24px",
  "32": "32px",
  "40": "40px",
  "48": "48px",
```

```
    "64": "64px"
  },
  "radius": {
    "sm": "8px",
    "md": "12px",
    "lg": "16px",
    "xl": "24px"
  },
  "stroke": {
    "hairline": "1px",
    "medium": "2px"
  },
  "shadow": {
    "sm": "0 1px 2px rgba(15,26,31,0.08)",
    "md": "0 8px 24px rgba(15,26,31,0.12)",
    "lg": "0 16px 40px rgba(15,26,31,0.14)"
  },
  "typography": {
    "fontFamily": {
      "sans": "Inter, ui-sans-serif, system-ui"
    },
    "size": {
      "12": "12px",
      "14": "14px",
      "16": "16px",
      "18": "18px",
      "20": "20px",
      "24": "24px",
      "32": "32px",
      "40": "40px"
    },
    "lineHeight": {
      "16": "16px",
      "20": "20px",
      "24": "24px",
      "28": "28px",
      "40": "40px",
      "48": "48px"
    }
  }
}
```

```
    },
    "weight": {
      "regular": "400",
      "medium": "500",
      "semibold": "600",
      "bold": "700"
    }
  },
  "component": {
    "topBar": {
      "height": "72px",
      "paddingX": "{space.24}",
      "paddingY": "{space.16}"
    },
    "contextRail": {
      "width": "360px",
      "padding": "{space.16}"
    },
    "actionDock": {
      "height": "72px",
      "paddingX": "{space.24}",
      "paddingY": "{space.12}"
    },
    "card": {
      "padding": "{space.16}",
      "radius": "{radius.lg}",
      "border": "{stroke.hairline}"
    },
    "button": {
      "height": "44px",
      "radius": "{radius.md}",
      "paddingX": "{space.16}"
    }
  }
}
```

3.3 CSS Variables (Drop-in)

```
:root {  
  /* Color primitives */  
  --navy-500: #1B3A4B;  
  --teal-500: #3A7B8A;  
  --green-500: #9AAF45;  
  --gray-900: #0F1A1F;  
  --gray-600: #5A6A72;  
  --gray-200: #D9E0E3;  
  --gray-100: #EEF2F3;  
  --white: #FFFFFF;  
  --red-500: #D64545;  
  --yellow-500: #F2C94C;  
  
  /* Semantic */  
  --bg-default: var(--gray-100);  
  --surface-primary: var(--white);  
  --surface-secondary: var(--gray-100);  
  --text-primary: var(--gray-900);  
  --text-secondary: var(--gray-600);  
  --border-subtle: var(--gray-200);  
  
  --status-ready: var(--green-500);  
  --status-atRisk: var(--yellow-500);  
  --status-notReady: var(--red-500);  
  
  /* Space scale */  
  --s0: 0;  
  --s2: 2px;  
  --s4: 4px;  
  --s8: 8px;  
  --s12: 12px;  
  --s16: 16px;  
  --s20: 20px;  
  --s24: 24px;  
  --s32: 32px;  
  --s40: 40px;  
  --s48: 48px;
```

```
--s64: 64px;

/* Radius */
--r-sm: 8px;
--r-md: 12px;
--r-lg: 16px;
--r-xl: 24px;

/* Shadow */
--shadow-sm: 0 1px 2px rgba(15,26,31,0.08);
--shadow-md: 0 8px 24px rgba(15,26,31,0.12);
--shadow-lg: 0 16px 40px rgba(15,26,31,0.14);

/* Component sizing */
--topbar-h: 72px;
--actiondock-h: 72px;
--contextrail-w: 360px;
}
```

4) FIGMA VARIABLES AND STYLES SETUP (WHAT YOUR DESIGNER DOES)

4.1 Figma Variables Collections

Create 3 collections:

Collection: **Core**

- space/8, space/12, space/16...
- radius/lg...
- stroke/1...

Collection: **Theme**

- color/bg/default

- `color/surface/primary`
- `color/text/primary`
- `color/border/subtle`
- `color/status/ready...`

Collection: **VerticalDictionary**

This is purely text tokens (or component properties) to swap nomenclature without redesign:

- `term/ecosystem` = Venue | City | Airport
- `term/event` = Game Day | Incident | Operational Day
- `term/readiness` = Game Ready | City Ready | Ops Ready

In Figma, implement as:

- Component text properties (preferred), or
 - Variables for strings (if supported in your current Figma setup), otherwise maintain a dictionary page.
-

5) RESPONSIVE RULES (DEV + FIGMA CONSISTENT)

Breakpoints

- **1440+**: Full layout (ContextRail visible)
- **1280**: ContextRail collapses to icon tabs + drawer
- **1024**: SideNav collapses to icon-only, ContextRail becomes drawer
- **768**: Mobile layout (separate spec later)

ContextRail Drawer Behavior

- Hidden by default
 - Opens from right
 - Width: `min(360px, 90vw)`
-

6) WHAT I NEED YOUR TEAM TO BUILD FIRST (MOST LEVERAGE)

If you build these components first, everything else becomes assembly:

1. **KPI Tile**
2. **Status Chip** (Ready, At Risk, Not Ready)
3. **Evidence Row**
4. **Action Button** (Primary, Secondary, Ghost)
5. **Rail Tabs**
6. **Card** (base container)

All with the tokens above.

If you want the next step, I can produce:

- A **token-to-component mapping table** (each component and which tokens it consumes)
- A **complete component inventory** (with variants and states)
- A **dev handoff contract** (naming rules, JSON keys, versioning) so engineering stays aligned as you scale across verticals.

a token to component mapping , map screens to data schema,

Below are two deliverables you can hand directly to design + engineering:

1. **Token-to-Component Mapping** (exact tokens each component consumes)
2. **Screen-to-Data Schema Mapping** (which objects, key fields, and relationships drive each template)

No methodology references. Engine-first. Vertical nomenclature handled via a dictionary layer.

1) Token-to-Component Mapping

Legend

- **Semantic tokens** drive consistent meaning across themes and verticals.
- **Primitive tokens** are fallback values (rarely used directly in components).

A. Foundations

Component	Tokens Used	Notes
AppShell Root	<code>color.bg.default</code> , <code>space.0</code>	Page background and reset padding
TopBar	<code>component.topBar.height</code> , <code>component.topBar.paddingX</code> , <code>component.topBar.paddingY</code> , <code>color.surface.primary</code> , <code>color.border.subtle</code> , <code>stroke.hairline</code> , <code>typography.*</code>	Global command frame
ContextRail	<code>component.contextRail.width</code> , <code>component.contextRail.padding</code> , <code>color.surface.primary</code> , <code>color.border.subtle</code> , <code>stroke.hairline</code>	Right rail persistent on desktop
ActionDock	<code>component.actionDock.height</code> , <code>component.actionDock.paddingX</code> , <code>component.actionDock.paddingY</code> , <code>color.surface.primary</code> , <code>color.border.subtle</code> , <code>stroke.hairline</code>	Always-on actions

B. Containers & Cards

Component	Tokens Used	Notes
Card Base	<code>component.card.padding</code> , <code>component.card.radius</code> , <code>stroke.hairline</code> , <code>color.surface.primary</code> , <code>color.border.subtle</code> , <code>shadow.sm</code>	Most UI blocks inherit from this
Panel (Large)	<code>space.24</code> , <code>radius.lg</code> , <code>color.surface.primary</code> , <code>shadow.md</code>	Viz panels, 3D viewport container
Section Header	<code>typography.size.18</code> , <code>typography.weight.semibold</code> , <code>color.text.primary</code> , <code>space.8</code>	Use consistently across screens

C. Metrics & Status

Component	Tokens Used	Notes
KPI Tile	<code>component.card.*</code> , <code>typography.size.24/32</code> , <code>typography.weight.semibold</code> , <code>color.text.primary</code> , <code>color.text.secondary</code>	Numeric-first
Readiness Badge	<code>radius.md</code> , <code>space.8</code> , <code>typography.size.12/14</code> , <code>color.status.ready/atRisk/notReady</code> , <code>color.text.inverse</code>	Status must be single source of truth
Risk Indicator (Pill)	<code>radius.xl</code> , <code>space.8</code> , <code>color.text.inverse</code> , <code>color.status.*</code>	Same status palette as readiness
SLA Timer	<code>typography.size.14/16</code> , <code>typography.weight.medium</code> , <code>color.text.primary</code> , <code>color.text.secondary</code> , <code>color.status.*</code>	Status color changes at thresholds

D. Actions & Inputs

Component	Tokens Used	Notes
Button Primary	<code>component.button.height</code> , <code>component.button.radius</code> , <code>component.button.paddingX</code> , <code>color.surface.inverse</code> , <code>color.text.inverse</code> , <code>shadow.sm</code>	Primary CTAs (Verify, Resolve)
Button Secondary	Same sizing tokens, <code>color.surface.secondary</code> , <code>color.text.primary</code> , <code>color.border.subtle</code>	For supporting actions
Button Ghost	Same sizing tokens, transparent fill, <code>color.text.accent</code>	Toolbar actions
Segmented Control	<code>radius.md</code> , <code>space.8/12</code> , <code>color.surface.secondary</code> , <code>color.border.subtle</code>	Tabs for ContextRail and view switches

Dropdown / Selector	<code>radius.md</code> , <code>space.12/16</code> , <code>color.surface.primary</code> , <code>color.border.subtle</code> , <code>typography.size.14</code>	Ecosystem selector, time horizon, role
---------------------	---	--

E. Tables, Lists, Evidence

Component	Tokens Used	Notes
Table Header	<code>color.surface.secondary</code> , <code>typography.size.12</code> , <code>color.text.secondary</code> , <code>space.12</code>	Must remain subtle
Table Row	<code>space.12</code> , <code>color.surface.primary</code> , <code>color.border.subtle</code>	Hover: adjust surface (optional)
Evidence Row	<code>space.12</code> , <code>radius.md</code> , <code>color.surface.primary</code> , <code>color.border.subtle</code> , <code>typography.size.14</code> , <code>color.text.secondary</code>	Immutable feel: no playful styling
Timeline Node	<code>color.border.subtle</code> , <code>color.text.secondary</code> , <code>space.8</code>	Evidence ledger timeline
Alert Chip	<code>radius.xl</code> , <code>space.8</code> , <code>color.status.*</code> , <code>color.text.inverse</code> , <code>typography.size.12</code>	Use chips sparingly

F. 3D / Spatial Controls

Component	Tokens Used	Notes
Layer Toggle	<code>segmented control tokens</code> , <code>color.text.primary</code> , <code>color.text.secondary</code>	Show layers: crowd, assets, incidents
Time Scrubber	<code>space.12</code> , <code>radius.md</code> , <code>color.surface.primary</code> , <code>shadow.sm</code> , <code>color.border.subtle</code>	Always floats above viewport
Legend	<code>typography.size.12</code> , <code>color.text.secondary</code> , <code>space.8</code>	For overlays and heatmaps

2) Screen-to-Data Schema Mapping

Canonical Engine Objects

These are your platform's "data spine." Every screen reads from the same objects.

Objects

- **Ecosystem**
- **Operation** (the active scope, like Game Day, Incident, Operational Day)
- **Stage** (time horizon and phase structure)
- **ReadinessCheck**
- **Task**
- **Asset**
- **Incident**
- **Evidence**
- **Score**
- **SLA**
- **Vendor**
- **Person** (roles, assignments)
- **Location** (zones, gates, sectors)
- **SensorSignal** (optional, for real-time and 3D overlays)

Relationships

- Ecosystem → has many Operations
- Operation → has many Stages
- Stage → has many ReadinessChecks, Tasks, Incidents, Scores
- Task → may reference Asset, Location, Vendor; produces Evidence
- Incident → produces Evidence; impacts Score; triggers SLA timers
- Vendor → owns tasks; provides compliance artifacts as Evidence
- SensorSignal → ties to Location/Asset/Incident in real time

A. Data Schema (Core Fields)

Use this as your baseline for API + DB + UI binding.

Ecosystem

- **ecosystem_id**
- **name**
- **vertical_type** (sports, civic, aviation)
- **geo** (optional)

- `timezone`
- `owner_org`
- `status`

Operation

- `operation_id`
- `ecosystem_id`
- `name` (ex: Week 6, Super Bowl, Citywide Drill)
- `start_time`, `end_time`
- `operation_type`
- `current_stage_id`
- `risk_level`

Stage

- `stage_id`
- `operation_id`
- `stage_key` (pre, in, post, renewal)
- `name` (vertical dictionary renders label)
- `start_time`, `end_time`
- `sequence`

ReadinessCheck

- `check_id`
- `stage_id`
- `category` (safety, access, comms, staffing)
- `requirement_text`
- `owner_person_id`
- `status` (not_ready, at_risk, ready, verified)
- `due_time`
- `evidence_required` (bool)

Task

- `task_id`
- `stage_id`
- `title`
- `description`
- `owner_person_id`

- `assigned_team`
- `status`
- `priority`
- `dependency_task_ids[]`
- `related_asset_ids[]`
- `related_location_ids[]`
- `sla_id` (optional)

Asset

- `asset_id`
- `ecosystem_id`
- `asset_type`
- `name`
- `status`
- `last_check_time`
- `location_id`
- `owner_vendor_id` (optional)

Incident

- `incident_id`
- `stage_id`
- `incident_type`
- `severity`
- `status` (open, contained, resolved)
- `reported_time`
- `location_id`
- `assigned_person_id`
- `related_tasks[]`
- `root_cause` (post-event)

Evidence

- `evidence_id`
- `linked_object_type` (task, incident, check, vendor)
- `linked_object_id`
- `timestamp`
- `source_system`
- `evidence_type` (photo, log, doc, attestation, sensor)
- `uri`

- `hash` (immutability)
- `verified_by_person_id`
- `verification_status`

Score

- `score_id`
- `stage_id`
- `score_type` (readiness, compliance, operations)
- `value` (0–100)
- `calculation_version`
- `updated_time`
- `drivers[ ]` (top contributors)

SLA

- `sla_id`
- `name`
- `thresholds` (warn, breach)
- `target_minutes`
- `breach_minutes`
- `metric_type`

Vendor

- `vendor_id`
- `name`
- `category`
- `status`
- `compliance_requirements[ ]`
- `validated_time`

Person

- `person_id`
- `name`
- `role`
- `org`
- `contact`
- `permissions[ ]`

Location

- `location_id`
- `ecosystem_id`
- `name`
- `location_type` (gate, zone, sector, terminal)
- `geo_boundary` (optional)

SensorSignal (optional)

- `signal_id`
- `timestamp`
- `signal_type` (crowd_density, door_state, camera_alert)
- `value`
- `location_id`
- `asset_id` (optional)

3) Screen Mapping (Templates → Objects & Queries)

Template T1: Command Dashboard

Uses

- Ecosystem, Operation, Stage
- Score, Incident, Task, SLA
- SensorSignal (optional), Location

Primary Queries

- `GET /operations/{id}/summary`
- `GET /stages/{id}/scores`
- `GET /stages/{id}/incidents?status=open`
- `GET /stages/{id}/tasks?status!=done&priority>=medium`
- `GET /signals/recent?operation_id=...` (optional)

UI Bindings

- KPI tiles: Score + counts of incidents/tasks
 - Alerts panel: Incidents + SLA breaches
 - Viz panel: Location overlays, signals, incidents
-

Template T2: Workflow Execution

Uses

- Stage, Task, SLA, Person, Evidence, Dependency graph
- Asset, Location

Primary Queries

- `GET /stages/{id}/workflow` (ordered tasks + dependencies)
- `GET /tasks/{id}`
- `GET /tasks/{id}/evidence`
- `GET /tasks/{id}/dependencies`

UI Bindings

- StepRail: Task list grouped by category or time window
 - WorkSurface: Task detail + action buttons
 - ContextRail: Evidence, dependencies, related incidents
-

Template T3: Readiness & Compliance Checklist

Uses

- ReadinessCheck, Evidence, Person, Vendor (if vendor-owned checks)
- Score (readiness score updates)

Primary Queries

- `GET /stages/{id}/checks?group_by=category`
- `GET /checks/{id}/evidence`
- `POST /checks/{id}/verify`

UI Bindings

- Checklist rows: requirement_text, owner, due_time, status
 - Evidence drawer: evidence list and verification
-

Template T4: Evidence Ledger

Uses

- Evidence (and linked object metadata)
- Person, Vendor, Source systems

Primary Queries

- `GET /operations/{id}/evidence?filters=...`
- `GET /evidence/{id}`
- `POST /evidence/{id}/verify`

UI Bindings

- Timeline: evidence timestamp ordered
 - Row: source_system, linked_object_type, verification_status
-

Template T5: After Action Insights

Uses

- Incident, Task, Score, SLA, Evidence, Location
- Aggregations and derived metrics

Primary Queries

- `GET /operations/{id}/analytics/bottlenecks`
- `GET /operations/{id}/analytics/sla-performance`
- `GET /operations/{id}/analytics/incident-trends`
- `GET /operations/{id}/report/summary`

UI Bindings

- Heatmaps: Location + density/incidents
 - Bottleneck cards: where delays happened, by time window
 - Training flags: repeated misses by team/role
-

Template T6: Certification & Scorecard

Uses

- Score, SLA, Vendor, Evidence, ReadinessCheck
- Certification status object (derived)

Primary Queries

- `GET /operations/{id}/certification/scorecard`
- `GET /vendors?operation_id=...&status!=validated`
- `GET /compliance/audit-trail?operation_id=...`

UI Bindings

- Benchmarks: score values, thresholds, pass/fail
 - Certification badge: derived state + renewal timeline
-

Template T7: 3D Spatial Command View

Uses

- Location, SensorSignal, Incident, Asset, Task
- Stage time scrubber

Primary Queries

- `GET /ecosystems/{id}/model` (3D model metadata)
- `GET /signals/recent?stage_id=...`
- `GET /incidents?stage_id=...`
- `GET /assets?stage_id=...`

UI Bindings

- Layer toggles = filter logic over signals/incidents/assets
 - Time scrubber = stage_time window on all queries
-

4) Vertical Nomenclature Layer (How screens stay consistent)

Implement as a dictionary returned by API or config:

Dictionary object

- `term.ecosystem`
- `term.operation`
- `term.stage`

- `term.readiness`
- `term.certification`
- `term.location`

Example usage

- UI renders “Ecosystem” label as `term.ecosystem` for that vertical.
 - Data schema stays the same. Only labels change.
-

What I can produce next (most valuable)

- A **component inventory** with variants/states and the exact API payloads each expects (perfect for dev handoff).
- A **JSON schema file** (OpenAPI-style) for the core objects above so engineering can implement quickly.
- A **sample dataset** for one vertical (NFL-style nomenclature) to populate Figma prototypes and demos.

Create vertical-specific screen copies Design motion rules for 3D transitions

Below are **production-ready vertical screen copies** plus **explicit 3D motion rules** you can hand to design, product, and engineering. This is written as **UI copy + interaction spec**, not marketing language. It will scale cleanly across demos, pilots, and regulated deployments.

No framework names. No abstraction leakage. Everything maps to the engine you already defined.

PART I — VERTICAL-SPECIFIC SCREEN COPY

(Same screens. Same data. Different language layer.)

Vertical 1: Elite Sports & Live Events

(NFL, leagues, venues, mega-events)

T1: Command Dashboard

Title: Game Operations Command

Subtitle: Live Readiness and Risk Overview

Primary KPIs

- Game Readiness Score
- Active Issues
- SLA Timers
- Gate & Zone Status
- Certification Status

Status Language

- Ready
- At Risk
- Not Ready
- Verified

Alerts Panel Copy

- “Ingress congestion detected at Gate C”
 - “Medical response SLA approaching threshold”
 - “Vendor credential validation pending”
-

T2: Workflow Execution

Title: Game Day Execution

Stage Labels

- Pre-Game
- In-Game
- Post-Game
- Renewal

Step Labels

- Verify Staff & Vendors
- Confirm Asset Readiness
- Monitor Crowd Flow
- Resolve Incidents
- Close & Certify

Action Buttons

- Verify Readiness
 - Escalate Issue
 - Mark Resolved
-

T3: Readiness Checklist

Title: League Standards Readiness

Checklist Groups

- Safety & Security
- Medical Coverage
- Access Control
- Communications
- Staffing

Row Labels

- Requirement
 - Owner
 - Status
 - Evidence
-

T4: Evidence Ledger

Title: Game Day Evidence Log

Column Headers

- Time
- Source
- Action
- Outcome
- Verified By

Empty State Copy

“All operational actions are logged automatically.”

T5: After Action Insights

Title: Post-Game Review

Insight Cards

- Bottlenecks Identified
 - SLA Misses
 - Staffing Load
 - Fan Experience Friction
-

T6: Certification

Title: League Certification Status

Badge Copy

“Game Ready. Certified.”

T7: 3D Command View

Title: Venue Live View

Layer Toggles

- Crowd Density
 - Medical Coverage
 - Security Zones
 - Asset Status
-
-

Vertical 2: Civic & Emergency Operations

(Cities, SEAR events, emergency management)

T1: Command Dashboard

Title: City Operations Command

Subtitle: Real-Time Incident Posture

Primary KPIs

- City Readiness Score
- Active Incidents
- Response SLAs

- Agency Status
- Duty of Care Status

Alerts Copy

- “Ingress congestion near perimeter sector 4”
 - “Fire response SLA nearing threshold”
 - “Inter-agency handoff pending”
-

T2: Workflow Execution

Title: Incident Operations

Stage Labels

- Preparedness
- Active Response
- Stabilization
- Recovery

Step Labels

- Verify Agency Readiness
 - Deploy Resources
 - Coordinate Response
 - Resolve Incident
 - Document Outcome
-

T3: Readiness Checklist

Title: ICS / NIMS Readiness

Checklist Groups

- Life Safety
 - Communications
 - Logistics
 - Command & Control
 - Mutual Aid
-

T4: Evidence Ledger

Title: Incident Evidence Log

Empty State Copy

“All response actions are time-stamped for audit and review.”

T5: After Action Insights

Title: After Action Review

Insight Cards

- Response Delays
 - Coordination Gaps
 - Resource Utilization
 - Training Opportunities
-

T6: Certification

Title: Duty of Care Status

Badge Copy

“Operationally Certified”

T7: 3D Command View

Title: Citywide Live View

Layer Toggles

- Incident Zones
 - Resource Deployment
 - Traffic Flow
 - Infrastructure Status
-
-

Vertical 3: Aviation & Airport Operations

T1: Command Dashboard

Title: Airport Operations Command
Subtitle: Airside and Landside Readiness

Primary KPIs

- Operational Readiness Score
 - Active Disruptions
 - Turnaround SLAs
 - Terminal Status
 - Compliance Status
-

T2: Workflow Execution

Title: Operational Day Execution

Stage Labels

- Pre-Ops
- Active Ops
- Recovery
- Renewal

Step Labels

- Verify Airside Readiness
 - Confirm Ground Assets
 - Monitor Passenger Flow
 - Resolve Disruptions
 - Certify Operations
-

T3: Readiness Checklist

Title: FAA / TSA Compliance Readiness

Checklist Groups

- Airside Safety
 - Passenger Screening
 - Communications
 - Staffing
 - Vendor Compliance
-

T4: Evidence Ledger

Title: Operations Evidence Log

T5: After Action Insights

Title: Operational Review

Insight Cards

- Gate Bottlenecks
 - Screening Delays
 - Staffing Imbalances
 - SLA Breaches
-

T6: Certification

Title: Compliance Validation

Badge Copy

“Ops Ready. Validated.”

T7: 3D Command View

Title: Airport Live View

Layer Toggles

- Passenger Density
 - Gate Utilization
 - Airside Assets
 - Security Zones
-

PART II — MOTION RULES FOR 3D & ADVANCED TRANSITIONS

These rules ensure the platform feels **calm, authoritative, and predictable**, not flashy.

1. CORE MOTION PRINCIPLES

Rule 1: Motion conveys state change, not decoration

If nothing meaningful changes, nothing animates.

Rule 2: Spatial continuity is sacred

Users must never “lose their place” in 3D space.

Rule 3: Time > Speed

Animations should feel deliberate, not fast.

2. GLOBAL MOTION TOKENS (DEV + DESIGN)

Token	Value
motion.fast	120ms
motion.standard	220ms
motion.slow	360ms
easing.standard	cubic-bezier(0.2, 0, 0, 1)
easing.exit	cubic-bezier(0.4, 0, 1, 1)

3. 3D VIEW TRANSITIONS (T7)

A. Entering 3D Command View

Trigger: Switch from Dashboard → 3D View
Motion

- Cross-fade dashboard (120ms)
- Camera eases into default altitude (360ms)
- Environment fades in from desaturated to full color

Purpose

Signals mode change without shock.

B. Camera Navigation

Rule

- No snap jumps.
- Always pan, then zoom.

Pan Duration: 220ms

Zoom Duration: 360ms

Easing: standard

Constraint

Max camera movement per interaction: 1 zone radius.

C. Layer Toggle (Crowd, Asset, Incident)

On

- Layer fades in (120ms)
- Overlay lines draw outward from origin (220ms)

Off

- Layer opacity fades to 0 (120ms)
 - No reverse drawing
-

D. Incident Highlight

Trigger: Click incident from alert list

Motion

1. Camera pans to incident zone (220ms)
2. Zone softly pulses once (360ms)
3. ContextRail opens (220ms)

Never

- Flash red
 - Shake
 - Bounce
-

E. Time Scrubber

Motion

- Scrubbing moves model state, not camera
- Entities fade in/out as time crosses thresholds (120ms)

Purpose

Shows evolution, not playback.

4. STATUS CHANGE MOTION (ALL SCREENS)

Readiness Status Change

- Color transition only
- 120ms fade
- No movement

SLA Breach

- Timer text color transitions
 - Subtle underline appears
 - No blinking
-

5. WORKFLOW STEP TRANSITIONS

Step Completion

- Step card border transitions to success color (120ms)
 - Checkmark fades in (120ms)
 - Next step subtly elevates (shadow increase only)
-

6. CONTEXT RAIL INTERACTIONS

Open

- Slide from right (220ms)
- Content fades in after 60ms delay

Close

- Fade content out (120ms)
 - Slide rail out (220ms)
-

7. WHAT WE NEVER DO (IMPORTANT)

- No bounce easing
- No elastic motion
- No parallax on critical ops screens
- No auto-rotating 3D views
- No attention-seeking animation

This is **operational intelligence**, not a consumer app.

FINAL DESIGN CHECK

If a user:

- Switches verticals
- Switches time horizons
- Switches from dashboard to 3D

They should feel:

- Oriented
- In control
- Calm

That is the brand.